

MDS-UPDRS Finger Tapping Analysis

The data provided below consists of extracted features (tap count, periods, and amplitudes) derived from a subject performing the MDS-UPDRS Item 3.4 Finger Tapping test. The subject was instructed to tap the index finger on the thumb 10 times as quickly and as largely as possible.

The objective is to score the subject according to MDS-UPDRS item 3.4: FINGER TAPPING based on these 10 consecutive finger taps. The testing instructions and scoring criteria for MDS-UPDRS item 3.4: FINGER TAPPING are as follows:

Instructions to the examiner:

Each hand is tested separately. Demonstrate the task, but do not continue to perform the task while the patient is being tested.

Instruct the patient to tap the index finger on the thumb 10 times as quickly AND as large as possible. Rate each side separately, evaluating speed, amplitude, hesitations, halts, and decrementing amplitude.

Scoring Criteria:

0 (Normal): No problems

1 (Slight): Any of the following:

- a) The regular rhythm is broken with one or two interruptions or hesitations of the tapping movement
- b) Slight slowing
- c) The amplitude decrements near the end of the 10 taps

2 (Mild): Any of the following:

- a) 3 to 5 interruptions during tapping
- b) Mild slowing
- c) The amplitude decrements midway through the 10-tap sequence

3 (Moderate): Any of the following:

- a) More than 5 interruptions during tapping or at least one longer arrest (freeze) in ongoing movement
- b) Moderate slowing
- c) The amplitude decrements starting after the 1st tap

4 (Severe): Cannot or can only barely perform the task due to slowing, interruptions, or decrements

Below are the extracted features including tap count, periods, and amplitudes. Please analyze this information and provide your score on the last line.

EXTRACTED FEATURES

Number of taps: 11

Tap periods (frames): [86, 78, 85, 77, 87, 73, 75, 72, 79, 65]

Tap amplitudes (pixels): [145.2067491544384, 159.20113064925135, 156.40012787718558, 150.60212481900777, 147.05441169852742, 158.3729774930054, 156.4736399525492, 149.5125412799876, 150.41608956491322, 154.01298646542764, 144.40221604947757]

GPT-5

Analysis of extracted features for MDS-UPDRS Item 3.4 (Finger Tapping)

Given data

- Number of taps recorded: 11 (expected: 10; likely includes an initial reference or an extra detected tap, but we can still assess the sequence quality).
- Tap periods (frames) for 10 intervals: [86, 78, 85, 77, 87, 73, 75, 72, 79, 65]
- Tap amplitudes (pixels) for 11 taps: [145.21, 159.20, 156.40, 150.60, 147.05, 158.37, 156.47, 149.51, 150.42, 154.01, 144.40]

Note: Periods are between-consecutive-tap intervals; amplitudes are per tap.

Rhythm and interruptions

- Periods fluctuate modestly (range 65–87 frames). There is no clear evidence of pauses or halts; differences between adjacent periods are small-to-moderate and do not suggest a freeze or prolonged arrest.
- Count of distinct “interruptions” (pronounced hesitations/halts) is not evident from the period data; no outlier interval that is dramatically larger than the others.

Interpretation: No obvious “3–5 interruptions” or “>5 interruptions” or a prolonged arrest.

Speed (slowing)

- Average periods appear fairly consistent with no progressive lengthening across the sequence. The last period (65) is actually among the shortest, which argues against slowing.
- This pattern does not fit mild or moderate slowing.

Interpretation: No clear slowing.

Amplitude and decrement

- Amplitudes are generally high and fairly stable with small within-sequence variability.
- Check for systematic decrement:
 - Early taps: 145 → 159 → 156 → 151 → 147
 - Mid taps: 158 → 156 → 150 → 150
 - Late taps: 154 → 144
- There is not a monotonic decline. After drops, amplitudes rebound (e.g., 147 → 158; 150 → 154). The final tap is the lowest (144), suggesting a slight end-effect, but the overall pattern does not show a consistent decrement starting after the 1st tap or even clearly by mid-sequence. If anything, there may be a slight decrement near the very end.

Interpretation: At most, a slight end-sequence decrement; not a mid-sequence or early-starting decrement.

Overall criterion mapping

- 0 (Normal): Would require no problems—here, there is a small end-tap amplitude dip and modest variability, which can be consistent with normal, but the explicit presence of a slight end decrement nudges toward 1.
- 1 (Slight): Fits “amplitude decrements near the end of the 10 taps” and no clear slowing or multiple interruptions.
- 2 (Mild) or higher: Not supported—no 3–5 interruptions, no mid-sequence decrement, no slowing, no freeze.

Final judgment: Slight abnormality due to minor end-sequence amplitude decrement without slowing or interruptions.

Score: 1
